Abhishek Lakhani

Applied AI Engineer – NLP & LLM Systems

lakhaniabhi.it@gmail.com | +49 176 5797 2691 | abhisheklakhani-it.github.io/Abhishek-Lakhani | github.com/abhisheklakhani-it

linkedin.com/in/abhishek-lakhani-4896271a6 | Germany

PROFESSIONAL SUMMARY

M.Sc. student (Automotive Software Engineering, TU Chemnitz) who builds and measures NLP and LLM systems in Python. Launched SmartVisual, a live LLM data analytics app, and built tested projects in hybrid RAG (nDCG@10 0.728 vs 0.567 for TF-IDF), LLM structured extraction and German DistilBERT fine-tuning with ONNX int8 deployment (1.46x faster, 4x smaller). Seeking working-student, thesis or junior roles in Applied AI. No visa sponsorship required | English (fluent), German (B1)

PROFESSIONAL EXPERIENCE

Machine Learning Intern, Technische Universität Chemnitz 06/2025 – 01/2026 | Chemnitz, Germany
- Built a privacy-focused image classification pipeline in TensorFlow with federated learning across decentralised Solid Pod data sources.
- Wrote modular Python tooling (structured logging, CSV result tracking) for reproducible experiments.

Software Development Intern, Wokontech IT Solution 10/2021 – 03/2022 | Surat, India
- Built and trained object detection pipelines on 5K+ images, improving robustness through data curation.
- Evaluated models using standard metrics and managed annotation workflows with CVAT.

Backend Development Intern, White Orange Software 12/2020 – 05/2021 | Surat, India
- Built and maintained backend features in Core PHP and tuned MySQL queries for performance.

PROJECTS

SmartVisual – AI Data Analytics App (own product, live) ,
React, TypeScript, Supabase (Postgres, Edge Functions), LLM API, Cloudflare Workers, GitHub Actions
- Built and launched a web app where users upload CSV/Excel files and ask questions in English or German to get KPIs, 18 chart types, maps and short written insights.
- Designed for correct numbers: the LLM only interprets the question and phrases the explanation, while every figure is computed in code over the full dataset; a 22-question evaluation set on 3 datasets runs in CI (every value within 0.01 of independently computed answers).

RAG Document Q&A Service + Retrieval Benchmark, Python, FastAPI, BM25, BGE Embeddings, FAISS, OCR, SSE, Docker
github.com/abhisheklakhani-it/rag-docqa | github.com/abhisheklakhani-it/retrieval-eval-bench
- Built a retrieval-augmented generation (RAG) service that answers questions on uploaded PDFs, including scanned PDFs via OCR, with cited answers streamed over Server-Sent Events.
- Benchmarked TF-IDF, BM25, dense and hybrid retrieval on BEIR SciFact with a train/test split and significance tests, then switched the service to hybrid BM25 + dense retrieval: nDCG@10 0.728 vs 0.567 for TF-IDF. A calibrated similarity floor rejects 87% of off-topic questions; 73 tests.

German Text Classifier + ONNX Deployment,
Hugging Face Transformers, PyTorch, Qdrant, ONNX Runtime, int8 Quantization, FastAPI, Docker
github.com/abhisheklakhani-it/german-text-classifier | github.com/abhisheklakhani-it/onnx-inference-optimization
- Fine-tuned distilbert-base-german-cased on 10kGNAD news (9 topics): 0.897 macro-F1 versus 0.845 for an embeddings + Qdrant kNN baseline; McNemar test $p < 0.001$.
- Exported the model to ONNX with graph fusion and int8 dynamic quantization: 1.46x PyTorch throughput, 4x smaller model (68 MB) for -0.3 points accuracy, served by FastAPI from a 147 MB Docker image.

LLM Structured Extractor , **LLM Structured Outputs, Pydantic v2, FastAPI**
- Extracts validated JSON (title, company, skills, salary, remote policy) from messy English and German job postings, with Pydantic schema validation and an automatic retry on validation errors.
- Evaluated field by field on a held-out set; error-driven prompt changes raised all-fields-correct from 0.40 to 0.67 (mean of 3 runs). Served by FastAPI, 43 automated tests.

More projects
Semantic Search API (Qdrant hybrid search, nDCG@10 0.737) · Test Bench Commissioning App (FastAPI, PostgreSQL, 137 tests) · LinkedIn Post Generator (LangChain) · Event Staff Scheduling Platform (Next.js, 330+ tests; private)

EDUCATION

Gujarat Technological University, B.E. in Computer Science Engineering 08/2017 – 07/2021 | Gujarat, India

Technische Universität Chemnitz, 04/2022 - expected 07/2027 | Chemnitz
Automotive Software Engineering (M.Sc.)

SKILLS

Programming: Python, SQL, JavaScript, TypeScript
LLM & NLP: LLMs, Prompt Engineering, Few-Shot Prompting, Structured Outputs, RAG, LangChain, Hugging Face Transformers, Fine-Tuning (DistilBERT), Sentence Embeddings, Text Classification, Information Retrieval, LLM APIs (Anthropic, Ollama)
Search & Vector DB: Qdrant, FAISS, BM25, TF-IDF, Hybrid Search (RRF), Cross-Encoder Reranking
Machine Learning: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Model Evaluation (nDCG, MRR, Recall@k, F1, Significance Tests)
MLOps & Backend: FastAPI, REST APIs, Pydantic, Docker, Docker Compose, ONNX Runtime, Quantization, pytest, GitHub Actions (CI), SQLAlchemy, PostgreSQL, Streamlit
Web & Deployment: React, Next.js, Tailwind CSS, Supabase, Prisma, Vitest, Playwright, Cloudflare Workers, Vercel, Render
Tools & Methods: Git, GitHub, Linux, n8n, CVAT, Agile/Scrum, Code Reviews
Cloud (Basics): Microsoft Azure | GCP | AWS